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25.

$\Delta G = \Delta H - T \Delta S$

At 300K

$\Delta G = 30000 \text{ J mol}^{-1} - 300 \text{ K} \times 100 \text{ J K}^{-1} \text{ mol}^{-1}$

$\Delta G = 0$

above 300 K ; ΔG will be negative and reaction becomes spontaneous.

I Keys to the short answer questions:

53.

SOLUTION :

Given :

$n = 2$ moles

$V_i = 500 \text{ ml} = 0.5 \text{ lit}$

$V_f = 2 \text{ lit}$

$T = 25^0C = 298 \text{ K}$

$w = -2.303 \text{ nRTlog} \left(\frac{V_f}{V_i}\right)$

$w = -2.303 \times 2 \times 8.314 \times 298 \times \log \left(\frac{2}{0.5}\right)$

$w = -2.303 \times 2 \times 8.314 \times 298 \times \log(4)$

$w = -2.303 \times 2 \times 8.314 \times 298 \times 0.6021$

$w = -6871 \text{ J}$

$w = -6.871 \text{ kJ}$

54.

SOLUTION :

Given :

$T_i = 298 \text{ K}$

$T_f = 298.45 \text{ K}$

$k = 2.5 \text{ kJ K}^{-1}$

$m = 3.5\text{g}$

$M_m = 28$

heat evolved $= k \Delta T$

$= k (T_f - T_i)$

$= 2.5 \text{ kJ K}^{-1} (298.45 - 298)\text{K}$

$= 1.125\text{kJ}$

$\Delta H_c = \frac{1.125}{3.5} \times 28 \text{ kJ mol}^{-1}$

$\Delta H_c = 9 \text{ kJ mol}^{-1}$

55.

SOLUTION :

Given :

$T_{sys} = 77^0C = (77 + 273) = 350 \text{ K}$

$T_{surr} = 33^0 C = (33 + 273) = 306 \text{ K}$

$q = 245 \text{ J}$

$\Delta S_{sys} = \frac{q}{T_{sys}} = \frac{-245}{350} = -0.7 \text{ JK}^{-1}$

$\Delta S_{surr} = \frac{q}{T_{sys}} = \frac{+245}{306} = +0.8 \text{ JK}^{-1}$

$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$

$\Delta S_{univ} = -0.7 \text{ JK}^{-1} + 0.8 \text{ JK}^{-1}$

$\Delta S_{univ} = 0.1 \text{ JK}^{-1}$